

Leeds Bus Tracker API Documentation

Document Info

- Project: COMP3011 Coursework 1 - Leeds Bus Tracker API
- API type: REST over HTTP
- Default local base URL: `http://localhost:3000/api`
- Production base URL: `https://cwk1.onrender.com/api`
- Content type: `application/json`

Authentication

Protected endpoints require a JWT access token in the Authorization header.

Header format:

```
Authorization: Bearer <token>
```

How to obtain token:

1. Register user: `POST /auth/register`
2. Login user: `POST /auth/login`
3. Read `token` from login response JSON
4. Include token in protected requests

Authentication-related errors:

- `401 Unauthorized`: missing token
- `403 Forbidden`: invalid or expired token

Endpoints

1) Register User

- Method: `POST`
- Path: `/auth/register`
- Auth required: No

Request body:

```
{
  "username": "alice",
  "password": "alice123"
}
```

Success response (`201`):

```
{
  "message": "User registered successfully.",
  "user": {
    "id": 1,
    "username": "alice"
  }
}
```

Error responses:

- 400 missing username/password
 - 409 username already exists
 - 500 server error
-

2) Login User

- Method: POST
- Path: /auth/login
- Auth required: No

Request body:

```
{
  "username": "alice",
  "password": "alice123"
}
```

Success response (200):

```
{
  "message": "Login successful.",
  "token": "<jwt-token>"
}
```

Error responses:

- 400 missing username/password
 - 401 invalid username or password
 - 500 server error
-

3) Get Stops (with filters + pagination)

- Method: GET
- Path: /stops

- Auth required: No
- Query params:
 - `atco_code` (optional)
 - `name` (optional, partial match)
 - `locality` (optional, partial match)
 - `page` (optional, default `1`)
 - `limit` (optional, default `20`, max `100`)

Example request:

```
GET /api/stops?name=Station&page=1&limit=20
```

Success response (`200`):

```
{
  "data": [
    {
      "atco_code": "450012345",
      "stop_name": "Leeds Station",
      "street": "New Station St",
      "indicator": "A",
      "latitude": 53.794,
      "longitude": -1.547,
      "locality": "Leeds"
    }
  ],
  "page": 1,
  "limit": 20,
  "total": 245
}
```

Error responses:

- `500` server error

4) Create Stop (protected)

- Method: `POST`
- Path: `/stops`
- Auth required: Yes

Request body:

```
{
  "atco_code": "TEST001",
  "stop_name": "Test Stop",
}
```

```
"street": "Test Street",
"indicator": "A",
"latitude": 53.8,
"longitude": -1.55,
"locality": "Leeds"
}
```

Success response (201):

```
{
  "atco_code": "TEST001",
  "stop_name": "Test Stop",
  "street": "Test Street",
  "indicator": "A",
  "latitude": 53.8,
  "longitude": -1.55,
  "locality": "Leeds"
}
```

Error responses:

- 400 required fields missing (atco_code, stop_name, latitude, longitude)
- 401 missing token
- 403 invalid/expired token
- 500 server error

5) Update Stop (protected)

- Method: PUT
- Path: /stops/:id
- Auth required: Yes
- Path param:
 - id: stop ATCO code

Request body (all fields optional):

```
{
  "stop_name": "Updated Stop Name",
  "street": "Updated Street",
  "indicator": "B",
  "latitude": 53.81,
  "longitude": -1.54,
  "locality": "Leeds"
}
```

Success response (200):

```
{
  "atco_code": "TEST001",
  "stop_name": "Updated Stop Name",
  "street": "Updated Street",
  "indicator": "B",
  "latitude": 53.81,
  "longitude": -1.54,
  "locality": "Leeds"
}
```

Error responses:

- 401 missing token
 - 403 invalid/expired token
 - 404 stop not found
 - 500 server error
-

6) Delete Stop (protected)

- Method: DELETE
- Path: /stops/:id
- Auth required: Yes
- Path param:
 - id: stop ATCO code

Success response (200):

```
{
  "atco_code": "TEST001",
  "stop_name": "Updated Stop Name",
  "street": "Updated Street",
  "indicator": "B",
  "latitude": 53.81,
  "longitude": -1.54,
  "locality": "Leeds"
}
```

Error responses:

- 401 missing token
 - 403 invalid/expired token
 - 404 stop not found
 - 500 server error
-

7) Get Arrival Logs (with filters + pagination)

- Method: `GET`
- Path: `/logs`
- Auth required: No
- Query params:
 - `stop_id` (optional)
 - `route_number` (optional)
 - `page` (optional, default `1`)
 - `limit` (optional, default `20`, max `100`)

Example request:

```
GET /api/logs?route_number=12&page=1&limit=20
```

Success response (`200`):

```
{
  "data": [
    {
      "id": 101,
      "stop_id": "450012345",
      "route_number": "12",
      "scheduled_time": "2026-03-07T08:30:00.000Z",
      "actual_time": "2026-03-07T08:34:00.000Z",
      "delay_minutes": 4,
      "status": "late"
    }
  ],
  "page": 1,
  "limit": 20,
  "total": 10000
}
```

Error responses:

- `500` server error

8) Create Arrival Log (protected)

- Method: `POST`
- Path: `/logs`
- Auth required: Yes

Request body:

```
{
  "stop_id": "450012345",
```

```
"route_number": "12",
"scheduled_time": "2026-03-07T08:30:00.000Z",
"actual_time": "2026-03-07T08:34:00.000Z"
}
```

Success response (201):

```
{
  "id": 101,
  "stop_id": "450012345",
  "route_number": "12",
  "scheduled_time": "2026-03-07T08:30:00.000Z",
  "actual_time": "2026-03-07T08:34:00.000Z",
  "delay_minutes": 4,
  "status": "late"
}
```

Error responses:

- 400 missing required fields (stop_id, route_number, scheduled_time)
- 401 missing token
- 403 invalid/expired token
- 500 server error

9) Update Arrival Log (protected)

- Method: PUT
- Path: /logs/:id
- Auth required: Yes
- Path param:
 - id: log ID

Request body (all fields optional):

```
{
  "actual_time": "2026-03-07T08:36:00.000Z"
}
```

Success response (200):

```
{
  "id": 101,
  "stop_id": "450012345",
  "route_number": "12",
  "scheduled_time": "2026-03-07T08:30:00.000Z",

```

```
"actual_time": "2026-03-07T08:36:00.000Z",
"delay_minutes": 6,
"status": "late"
}
```

Error responses:

- 401 missing token
 - 403 invalid/expired token
 - 404 log not found
 - 500 server error
-

10) Delete Arrival Log (protected)

- Method: DELETE
- Path: /logs/:id
- Auth required: Yes
- Path param:
 - id: log ID

Success response (200):

```
{
  "message": "Log deleted"
}
```

Error responses:

- 401 missing token
 - 403 invalid/expired token
 - 404 log not found
 - 500 server error
-

11) Reliability by Stop

- Method: GET
- Path: /reliability/:atco_code
- Auth required: No

Success response (200):

```
{
  "atco_code": "450012345",
  "stats": {
    "total_arrivals": "120",
    "avg_delay": "1.80",
  }
}
```

```
    "early_count": "25",
    "late_count": "40",
    "cancelled_count": "5"
  }
}
```

Error responses:

- 404 stop not found or no arrivals
 - 500 server error
-

12) Reliability by Route

- Method: GET
- Path: /reliability/route/:route_number
- Auth required: No
- Query params:
 - stop_id (optional)

Example request:

```
GET /api/reliability/route/12?stop_id=450012345
```

Success response (200):

```
{
  "route_number": "12",
  "stop_id": "450012345",
  "stats": {
    "total_arrivals": "32",
    "avg_delay": "2.10",
    "early_count": "5",
    "late_count": "12",
    "cancelled_count": "2"
  }
}
```

Error responses:

- 404 route not found or no arrivals
- 500 server error

Postman / Testing Notes

You can test endpoints in Postman using:

- Base URL: <http://localhost:3000/api>

- For protected routes, set **Authorization** -> **Bearer Token** with the JWT from login.